SIH 2026 — SIH26084

Convective Scale Nowcasting for Thunderstorms, Hail & Cloudbursts (0–6 hr)

COMPLETE TECHNICAL PROJECT GUIDE

Item	Details
Organization	Ministry of Earth Sciences (MoES)
Department	National Centre for Medium Range Weather Forecasting (NCMRWF)
Category	Software
Theme	Disaster Management
Forecast horizon	0–6 hours
Target spatial scale	1–3 km

Practical blueprint: data → ML → backend → GIS → evaluation → deployment

1. Problem in Simple Language

Severe thunderstorms, hail, downburst winds and cloudbursts can develop very quickly and affect small areas. SIH26084 asks for a real-time system that can watch these events, identify early development and provide a short-term forecast for the next 0–6 hours at about 1–3 km spatial detail.

The official problem asks for multi-source data fusion using Doppler Weather Radar, INSAT-3D/3DR satellite imagery and ground-based lightning observations. It also asks for gridded forecasts of lightning strike density, hail probability, downburst velocity and cloudburst thresholds, shown through an interactive GIS dashboard with storm-arrival countdowns.

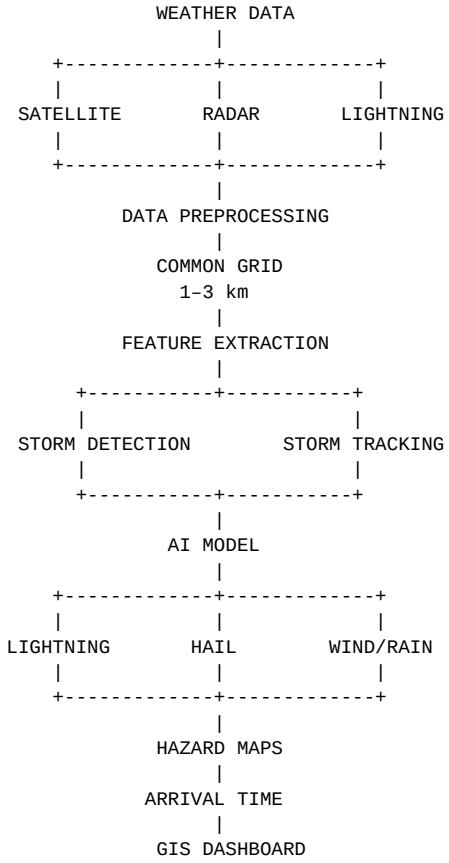
In practical terms, your system should:

- watch weather observations
- detect developing convection
- track storm movement
- estimate localized hazards
- show everything clearly on a map

2. Official Scope

Requirement	What it means
0–6 hour nowcast	Near-term forecast, not a multi-day forecast.
1–3 km	Localized grid-based hazard information.
Multi-source fusion	Combine observations from different sources.
Early initiation	Identify storm development before the mature stage.
Lightning density	Spatial lightning activity, not only yes/no.
Hail probability	Probability field.
Downburst velocity	Estimated damaging wind strength.
Cloudburst threshold	Extreme-rainfall risk/threshold output.
GIS dashboard	Interactive spatial visualization.
Arrival countdown	Estimated time to a selected location.

3. Overall Architecture



Build it in this order: satellite pipeline → detection → tracking → baseline → AI → APIs → dashboard → advanced radar/lightning fusion.

4. Data Sources

INSAT-3DR is a strong starting point. MOSDAC/ISRO documents six imager bands: VIS, SWIR, MIR, WV, TIR-1 and TIR-2. Nominal sub-satellite-point resolutions are 1 km for VIS/SWIR, 4 km for MIR/TIR-1/TIR-2 and 8 km for WV.

Source	Useful information	Role
INSAT-3D/3DR	Thermal/IR, visible/SWIR	Cloud development, temperature, motion
DWR	Reflectivity + velocity	Storm structure and movement
Lightning	Time + location	Lightning activity/density
Derived products	QPE, cloud motion etc.	Additional features

Important: do not say every INSAT channel is 1 km. The thermal TIR channels are 4 km; your 1–3 km output is an analysis/prediction grid, not the native resolution of every input channel.

5. Satellite Processing

```
HDF / GeoTIFF
      |
Read metadata + geolocation
      |
Extract required bands
      |
Crop selected region
      |
Reproject / resample
      |
Create cloud / temperature features
      |
Calculate temporal changes
      |
Save processed arrays
      |
ML pipeline
```

Useful features:

- cloud-top brightness temperature
- cloud-top cooling rate
- cloud area/growth
- cloud texture and gradients
- cloud motion
- rainfall/cloud-motion derived products where appropriate

Cloud-top cooling rate can be represented as CTCR = (T_current – T_previous) / time_difference. Treat it as a feature; choose thresholds using data/literature and validation.

6. Radar Processing

Create a radar adapter that converts an authorized radar product into the same common grid used by the satellite and lightning pipelines.

```
Radar product
      |
Read reflectivity / velocity
      |
Quality control
      |
Georeference
      |
Resample to common grid
      |
Create features
      |
Fuse with other sources
```

- reflectivity
- radial velocity
- maximum reflectivity
- rainfall intensity/accumulation where available
- storm-cell movement

Do not make raw radar decoding your first task. Use documented/authorized products and keep the adapter modular.

7. Lightning Processing

Store at least timestamp and geographic position for each observation. Aggregate observations into the common grid and forecast time window.

```
Lightning observations
      |
timestamp, latitude, longitude
      |
Spatial binning
      |
Count strikes per cell/time window
      |
Lightning-density feature
```

The SIH output is lightning strike density. A simple yes/no lightning layer is not the complete requirement.

Verify lawful access to live lightning data before promising a real-time multi-source system. Never fabricate observations.

8. Common Grid & Data Fusion

Radar, satellite and lightning data have different spatial resolutions, projections and update times. Create a common 1–3 km analysis grid.

```
Region
  |
Define grid
  |
Convert coordinates
  |
Resample where appropriate
  |
Attach time/source
  |
Create fused feature tensor
```

Example ML tensor: [batch, time, channels, height, width]. Example [8, 6, 5, 128, 128] means 8 samples, 6 historical frames, 5 channels and a 128×128 grid.

Resampling does not create new physical information. Keep native-resolution metadata for every source.

9. Storm Detection & Tracking

Before forecasting severity, answer: where is the storm?

- Start with cloud temperature, cooling rate and cloud growth.
- Use connected components/segmentation to identify candidate storm regions.
- Track cells across frames.
- Estimate direction and speed.
- Later replace the rule-based detector with CNN/U-Net if useful.

Optical flow can provide an early storm-motion estimate without needing a complex learned tracker.

10. Baseline Forecast

A strong SIH project should compare its AI model with simple baselines.

```
Persistence:
Current field -> assume it stays similar.

Advection:
Current field -> estimate motion -> move it forward.
```

Compare the AI against these baselines at 30, 60, 120 minutes and later horizons supported by your dataset. If your AI does not beat the baseline, the sophistication of the model is not enough.

11. AI / ML Model

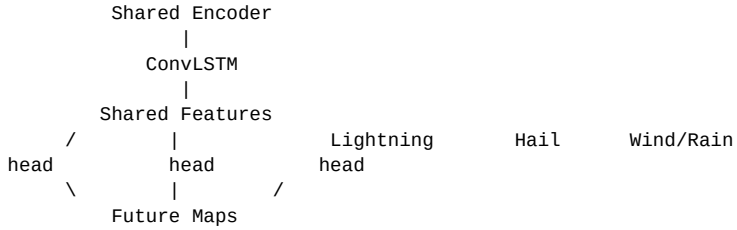
Recommended first serious model: CNN + ConvLSTM.

```
Historical frames
       |
      CNN
       |
Spatial features
       |
    ConvLSTM
       |
Temporal features
       |
Decoder / prediction heads
       |
Future hazard maps
```

The problem is spatial and temporal. CNN learns spatial patterns; ConvLSTM learns how those patterns evolve across time.

Start with a small region and a manageable number of channels. Scale only after the complete pipeline works.

12. Multi-Task Hazard Model



Output	Target
Lightning	Strike density
Hail	P(hail)
Downburst	Velocity
Cloudburst	Rainfall / threshold exceedance

If a target cannot be reliably labelled and validated for your chosen region, mark that component experimental instead of inventing labels.

13. Dataset Construction

```
Input:
T-120, T-90, T-60, T-30, T0

Target:
T+30, T+60, T+90, T+120 ...
```

- split by storm event/date, not random images
- keep validation events separate
- keep final test events unseen
- store timestamp/source for every sample
- store missing-data masks

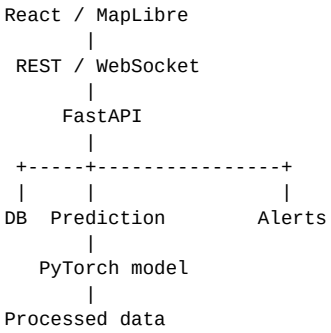
Randomly splitting frames from the same storm can make the model look artificially strong. Event-based splitting is much safer.

14. Evaluation

Metric	Purpose
CSI	Categorical forecast skill.
POD	How many observed events were detected.
FAR	How many forecast events were false alarms.
F1	Classification/segmentation balance.
MAE / RMSE	Continuous outputs such as wind/rain.
Brier score	Probabilistic forecasts.
Spatial similarity	Compare forecast and observed maps.

Always report the forecast horizon. Never put made-up numbers in the results table.

15. Backend Architecture



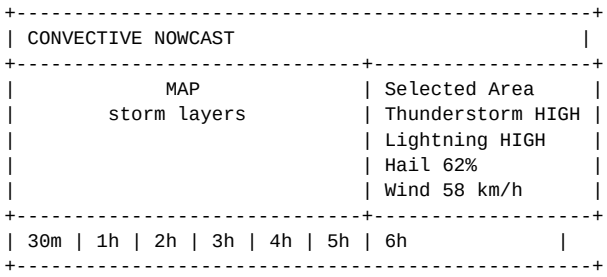
Endpoint	Purpose
GET /api/storms/current	Current storm cells and movement
GET /api/forecast	Forecast for a location
GET /api/hazards	GeoJSON hazard layers
GET /api/events	Historical replay events
WS /api/live	Live prediction updates

16. Database Design

Table	Important fields
satellite_observations	timestamp, band, geometry, value/path
radar_observations	timestamp, reflectivity, velocity, geometry
lightning_observations	timestamp, latitude, longitude, polarity
storm_cells	storm_id, timestamp, centroid, speed, direction, area
forecast_grid	forecast_time, geometry, hazard values
alerts	location, severity, arrival_time
model_runs	model_version, run_time, metrics

Use PostgreSQL + PostGIS for spatial queries and map-ready data.

17. GIS Dashboard



- storm cells
- lightning density
- hail probability
- downburst/wind
- heavy rain/cloudburst
- cloud temperature
- storm movement
- observed vs forecast

18. Storm Arrival Countdown

For a selected location, estimate when the storm is expected to reach it.

```
distance = D
storm speed = V

arrival_time ≈ D / V
```

Example: 45 km distance at 45 km/h gives a simple estimate of about one hour. In the real system, update the estimate whenever the storm track changes.

```
MEERUT
Storm distance: 42 km
Direction: NE
Estimated arrival: 49 min
Last update: 18:32
```

Show this as a model estimate, not a guarantee.

19. Real-Time Processing

```
New observation
      |
Data ingestion
      |
Queue / worker
      |
Preprocessing
      |
Feature extraction
      |
ML inference
      |
Post-processing
      |
PostGIS
      |
WebSocket
      |
Dashboard update
```

- FastAPI for APIs
- Redis + Celery when background workload grows
- PostgreSQL/PostGIS
- WebSocket for live updates
- Docker for reproducible deployment

20. Recommended Technology Stack

Layer	Tools
Language	Python
ML	PyTorch
Image processing	OpenCV
Satellite/GIS	xarray, h5py, rasterio, GeoPandas, Shapely, pyproj
Backend	FastAPI
Database	PostgreSQL + PostGIS
Frontend	React
Map	MapLibre or Leaflet
Queue	Redis + Celery (optional)
Deployment	Docker + cloud backend + web hosting

21. GitHub Structure

```
SIH26084/
■■■ frontend/
■   ■■■ src/  components/  maps/  pages/
■■■ backend/
■   ■■■ app/
■       ■■■ main.py  api/  models/  services/  schemas/
■■■ ml/
■   ■■■ preprocessing/  features/  models/
■   ■■■ training/  evaluation/
■■■ data/
■   ■■■ raw/  processed/  samples/
■■■ notebooks/
■■■ scripts/
■■■ docker/
■■■ requirements.txt
■■■ README.md
```

Keep restricted/large raw datasets out of GitHub. Store metadata and download instructions instead.

22. Team Division — 5 People

Role	Main work
Team Lead	Architecture, integration, backend coordination, demo
ML	CNN/ConvLSTM, training, validation
Data	MOSDAC data, HDF/GeoTIFF, preprocessing, grid
Frontend/GIS	React, map, dashboard
Backend/DevOps	FastAPI, PostGIS, WebSocket, deployment

23. Development Roadmap

Phase	Goal
1	Verify data access; load one historical event.
2	Preprocess satellite data and build the grid.
3	Detect convective regions.
4	Track storm speed/direction.
5	Build persistence + optical-flow baseline.
6	Train CNN/ConvLSTM.
7	Build FastAPI + database.
8	Build GIS dashboard.
9	Add replay + arrival clock.
10	Evaluate against baseline and document results.

24. MVP vs Advanced

Priority	Features
MUST HAVE	INSAT ingestion, preprocessing, storm detection/tracking, baseline, GIS map, historical replay
SHOULD HAVE	ConvLSTM, rainfall risk, lightning layer, arrival countdown
ADVANCED	Radar fusion, live lightning feed, hail probability, downburst velocity, full multi-hazard 0–6 hour model

Finish the working MVP before adding advanced modules.

25. Risks & Solutions

Risk	Response
Data access	Verify access before promising live radar/lightning.
Missing labels	Start with defensible targets; mark experimental outputs.
Different resolutions	Preserve native metadata and define a common grid.
Overfitting	Split by event/date.
Fake accuracy	Report only held-out test results.
Overbuilding	Finish satellite-first MVP first.
UI-first development	Build data/model pipeline before polishing UI.

26. What Not to Claim

- Do not claim 95% accuracy without test evidence.
- Do not claim live IMD/radar access unless authorized.
- Do not present simulated radar/lightning as real observations.
- Do not call an arrival estimate guaranteed.
- Do not call a resampled 1 km grid the native resolution of 4 km thermal imagery.
- Do not claim operational hail/downburst reliability without labels and validation.

Recommended wording: “Our current prototype is satellite-first and has modular interfaces for authorized radar and lightning feeds. We clearly separate observed, simulated and predicted information.”

27. First 48 Hours

Time	Task
0–4 hr	Repository, environment and folders.
4–10 hr	Investigate MOSDAC access; select one historical event.
10–18 hr	Read an INSAT file and plot bands.
18–28 hr	Geolocation, crop and common grid.
28–36 hr	Compare frames; visualize movement/cooling.
36–48 hr	Simple storm detector + first processed dataset.

Do not start with ConvLSTM. First prove that you can reliably read, visualize and preprocess the data.

28. SIH Demo Story

```
1. Select a historical storm event.
2. Start replay.
3. Show normal conditions.
4. Show rapid cloud development.
5. Detect convective initiation.
6. Show storm movement.
7. Run short-term forecast.
8. Turn on hazard layers.
9. Select a location.
10. Show estimated arrival time.
11. Compare forecast with actual event.
12. Show evaluation metrics.
```

The message to judges: this is not only a weather map. It is a pipeline that turns observations into storm detection, movement prediction, hazard information and an understandable early-warning view.

29. Final Project Definition

Project: A real-time GIS-based convective nowcasting platform that combines high-frequency meteorological observations, detects early storm development, tracks movement and produces short-term hazard information for the next 0–6 hours.

Core MVP: INSAT-based convective development detection + storm tracking + short-term forecast + GIS dashboard.

Advanced vision: Multi-source fusion of INSAT, Doppler radar and lightning observations with multi-hazard prediction.

Engineering principle: Build a working satellite-first pipeline first; add authorized sources and additional hazard heads only after the foundation is validated.

30. Official References

SIH26084: SIH 2026 problem listing — Ministry of Earth Sciences / NCMRWF. Official scope: 0–6 hour convective nowcasting, 1–3 km scale, multi-source fusion and GIS dashboard.

MOSDAC / ISRO: INSAT-3DR payloads and data products. Useful for channel information, native resolutions and derived meteorological products such as cloud-motion and precipitation products.

IMD: Radar/forecasting documentation for radar products such as reflectivity and velocity.

URLs for verification:

https://sih2026.vuce.in/ps/SIH26084

https://mosdac.gov.in/insat-3dr-payloads

https://www.mosdac.gov.in/insat-3dr-data-products

https://mausam.imd.gov.in/imd_latest/contents/pdf/forecasting_sop.pdf

Use the official SIH problem statement as the authority for the final submission requirements.